

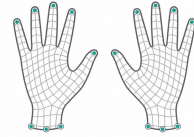
Training-Ready Annotation Spectrum

representation

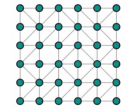
semantic role

model need

1 Dense MANO State / Action



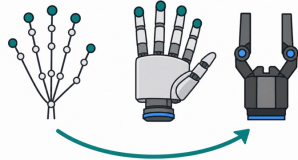
fine-grained hand prior



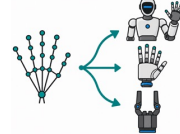
dense supervision



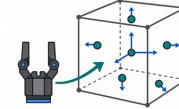
2 Robot-Facing Interfaces



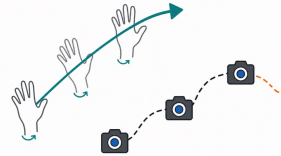
bridge to robot embodiments



normalized action space



3 Hand + Camera Dynamics



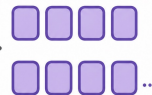
egocentric controllable change



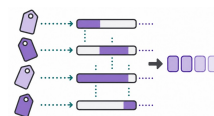
action-conditioned video dynamics



4 Latent / Weak Action



relaxed explicit labels



guided controllability



high

explicit supervision density

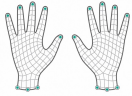
low

Open-AoE Segment

RGB Video



MANO Hands



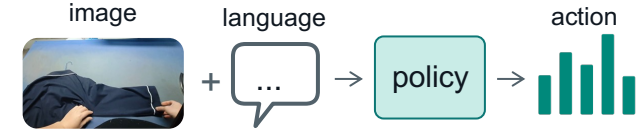
Camera Motion



Action Text

Smooth the sleeve

A VLA Policies



stable semantics

normalizable actions

VITRA

GR00T N1.7

H-RDT

SmolVLA

$\pi 0.5$

ACT

Diffusion Policy

...

B World Action Models



temporal action labels

camera-aware dynamics

DreamZero

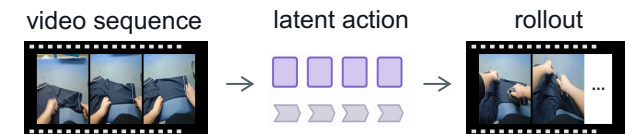
Lingbot-VA

Ctrl-World

iVideoGPT

...

C World Models



guided controllability

weak supervision

DreamDojo

AdaWorld

LAOM

GenieRedux

...